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SBEU 3213 FIELD ASTRONOMY

SECTION EB1

CASE STUDY 1: Why Does the Shadow Length Change Every Day?

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GROUP

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“We declare that this case study is our own work and that all group members contributed to the completion of this assignment.”

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1.0 Introduction

Field astronomy, the observational study of celestial phenomena through direct engagement with the natural environment, has profoundly shaped human civilization since antiquity. Unlike theoretical astronomy conducted through advanced instrumentation in controlled settings, field astronomy emphasizes accessible, ground-based observations that reveal fundamental astronomical principles through simple yet systematic methods (Krupp, 2003). These practices have historically underpinned essential aspects of daily life, from precise timekeeping systems to reliable navigation techniques that enabled global exploration.

The practical applications of field astronomy remain evident in contemporary society. Modern timekeeping, despite relying on atomic clocks, is fundamentally rooted in solar observations that defined the day-night cycle and established time zones based on longitudinal position relative to the Sun's apparent motion (McCarthy & Seidelmann, 2018). Similarly, celestial navigation—though largely supplanted by Global Positioning System (GPS) technology—continues to serve as a critical backup system for maritime and aviation operations, with training in astronomical position-finding remaining mandatory for commercial pilots and naval officers (Cutler, 2003).

One of the most fundamental yet scientifically rich phenomena observable through field astronomy is the shadow cast by a vertical object, known as a gnomon. Daily observations reveal that shadow length and direction vary systematically throughout the day, reaching minimum length at solar noon when the Sun achieves its highest altitude above the horizon (Rohr, 1996). More intriguingly, careful measurement demonstrates that the *shortest* shadow length at solar noon varies considerably throughout the year at any given location. This seemingly simple observation encapsulates profound astronomical principles: Earth's axial tilt, its orbital revolution around the Sun, and the resulting seasonal variation in solar declination.

This report examines the shadow phenomenon through systematic field observations, investigating why a gnomon casts shadows of different lengths at the same clock time on different days of the year. By connecting empirical observations with established astronomical theory, we demonstrate how this accessible phenomenon illuminates Earth's relationship with the Sun and its practical implications for both historical and modern applications.

2.0 Description of the Case Study Issue

This case study focuses on a simple observation. A fixed vertical stick (gnomon) does not cast the same shadow length at the same clock time on different days, even when the stick, ground surface, and observation point are unchanged. In a school setting, students may mark the tip of the shadow at a chosen clock time across multiple days and notice that the shadow endpoint shifts position and the shadow length grows or shrinks. The change is gradual day-to-day, but it becomes obvious over a few weeks and is most striking when comparing observations separated by months.

The cause is not measurement error, it is that clock time does not uniquely determine the Sun's position in the sky across the year. At a fixed location, the Sun's apparent path changes through the seasons because Earth's rotation axis is tilted relative to its orbital plane, causing the Sun's declination to vary over the year (Greenwood, 2022). Since the shadow length is controlled by solar elevation angle, even a small seasonal change in solar elevation at the same nominal time will change the shadow length (shorter shadows when the Sun is higher; longer shadows when it is lower) (Wang, 2017).

In practice, this issue occurs every clear day and can be observed anywhere on Earth, though the magnitude of the change depends strongly on latitude. Locations farther from the equator typically experience larger seasonal swings in midday solar altitude and therefore larger seasonal differences in shadow length (Lee et al., 2025). Additionally, even if students are careful to observe at the same clock time, the Sun's apparent motion relative to mean clock time varies across the year (the equation of time effect), which can add a smaller but real shift in solar position at the same time.

So the core problem is: why does a fixed object produce different shadow lengths at the same clock time on different dates? It is an astronomy problem because the shadow is effectively a "Sun position sensor". It directly reflects Earth–Sun geometry (apparent solar motion and seasonal solar declination) rather than anything changing about the stick.

3.0 Relevant Astronomy Concepts

Understanding shadow variation requires comprehension of three interconnected astronomical phenomena: the Sun's apparent daily motion across the celestial sphere, Earth's axial tilt relative to its orbital plane, and the resulting seasonal changes in solar declination.

3.1 The Sun's Apparent Motion

From an Earth-based observational perspective, the Sun appears to traverse the sky along a predictable arc from east to west, a phenomenon resulting from Earth's rotation on its axis (Karttunen et al., 2017). This apparent solar motion reaches its highest point at solar noon—defined not by clock time but by the Sun's transit across the local meridian, the imaginary north-south line passing through the observer's zenith (Duffett-Smith & Zwart, 2011). At solar noon, the Sun achieves its maximum altitude angle above the horizon for that particular day, consequently producing the shortest shadow length for any vertical object.

The altitude angle of the Sun at solar noon, however, is not constant throughout the year. This variation stems directly from Earth's orbital geometry and axial orientation (Milham, 1945). The relationship between solar altitude (h), observer latitude (ϕ), and solar declination (δ) at solar noon is expressed mathematically as:

$$h = 90^\circ - |\phi - \delta|$$

This equation demonstrates that solar noon altitude depends critically on both the observer's geographical position and the Sun's declination, which varies seasonally (Meeus, 1998).

3.2 Earth's Axial Tilt

Earth's rotational axis maintains a tilt of approximately 23.5° relative to the perpendicular of its orbital plane around the Sun, a geometric relationship known as obliquity of the ecliptic (Berger, 1976). This axial tilt remains nearly fixed in space throughout Earth's annual orbit, pointing toward Polaris (the North Star) regardless of Earth's position relative to the Sun. Consequently, the angle at which sunlight strikes any given latitude changes systematically as Earth progresses through its orbit (Roy & Clarke, 2003).

During the June solstice (approximately June 21), Earth's Northern Hemisphere tilts maximally toward the Sun, resulting in the Sun appearing at its highest point in northern skies and lowest in southern skies. Conversely, at the December solstice (approximately December 21), the Southern Hemisphere receives this maximum tilt toward the Sun (Espenak, 2007). The equinoxes (approximately March 20 and September 22) represent transitional moments when Earth's axis tilts neither toward nor away from the Sun, and the Sun crosses the celestial equator.

This axial tilt is the fundamental cause of Earth's seasons and the primary driver of shadow length variation throughout the year (Wolfson, 2020). Without this tilt, the Sun would rise and set at the same points on the horizon year-round, and shadows at solar noon would maintain constant length at any given location.

3.3 Seasonal Solar Declination

Solar declination (δ) represents the angular position of the Sun north or south of the celestial equator, measured in degrees (Smart, 1977). This value oscillates between approximately $+23.5^\circ$ (at the June solstice when the Sun appears directly overhead at the Tropic of Cancer, 23.5°N) and -23.5° (at the December solstice when the Sun appears directly overhead at the Tropic of Capricorn, 23.5°S). At the equinoxes, solar declination equals 0° , positioning the Sun directly above Earth's equator (Astronomical Almanac, 2024).

For observers at mid-latitudes in the Northern Hemisphere, such as Johor Bahru, Malaysia (approximately 1.5°N), solar declination variations produce measurable changes in shadow length throughout the year. When the Sun's declination is positive (Northern Hemisphere summer), the Sun rises higher in northern skies, casting shorter noon shadows. When declination is negative (Northern Hemisphere winter), the Sun's lower altitude produces longer noon shadows (Green, 1985).

The relationship between shadow length (L), gnomon height (H), and solar altitude (h) follows basic trigonometry:

$$L = H / \tan(h)$$

This equation reveals an inverse relationship: as solar altitude increases, shadow length decreases (Waugh, 1973). Because solar altitude at noon varies with declination, and declination changes seasonally due to Earth's axial tilt, shadow length at solar noon must similarly vary throughout the year.

3.4 Integration of Concepts

The interplay between Earth's axial tilt, orbital revolution, and resulting declination changes creates the observed shadow phenomenon. At locations near the equator like Johor Bahru (1.5°N), the Sun passes nearly overhead during equinoxes when solar declination approaches the observer's latitude, producing exceptionally short shadows (Mayall & Mayall, 1994). During solstices, when declination reaches its extremes ($\pm 23.5^\circ$), the Sun's path lies farther from the zenith, creating longer noon shadows even at this low-latitude location.

For mid-latitude observers in temperate regions, these variations become even more pronounced. At 40°N latitude, the solar altitude at noon ranges from approximately 73° at the June solstice (short shadows) to only 26° at the December solstice (long shadows)—a dramatic difference of 47° that produces shadow length variations of nearly 300% (Aveni, 2008).

This astronomical framework provides the theoretical foundation for understanding field observations of shadow variation, demonstrating how Earth's geometry relative to the Sun governs one of nature's most accessible yet scientifically significant phenomena.

4.0 Investigation Method

This investigation was conducted as a desk-based study using secondary sources. The aim was to explain why a fixed gnomon can produce different shadow lengths at the same clock time on different dates, by synthesising established astronomical theory with standard shadow-geometry relationships. The study relied on:

1. Peer-reviewed and educational literature describing seasonal Sun-angle effects and shadow-length behaviour (e.g., Greenwood, 2022; Wang, 2017)
2. Authoritative astronomy references and ephemeris material (e.g., Astronomical Almanac, 2024; Meeus, 1998; Duffett-Smith & Zwart, 2011) to support definitions and core relationships used in the explanation.

Analytical Procedure:

1. Define the controlling variables for shadow length, which include solar altitude and its dependence on observer latitude and solar declination, including the solar-noon relationship, $h = 90^\circ - |\phi - \delta|$.
2. Link solar altitude to shadow length using the standard gnomon geometry $L = \frac{H}{\tan(h)}$, so that any seasonal change in h directly implies a change in shadow length L for constant gnomon height H .
3. Account for “same clock time” effects by incorporating the concept that clock time does not uniquely fix the Sun’s apparent position across the year, due to seasonal changes in declination and the Equation of Time shifting apparent solar time relative to mean clock time.
4. Explanations and example trends reported across sources were cross-checked for agreement with the above relationships, and then integrated into a coherent causal narrative.

Because no primary measurements were collected, the “observations” discussed in the analysis refer to well-established, documented shadow behaviours described in the literature and explained via standard astronomical geometry. The method therefore emphasizes conceptual validity and source triangulation, rather than local-site measurement accuracy. In addition, the project schedule limited what could realistically be done as the time window was insufficient to repeat field observations across enough dates to capture meaningful seasonal variation, so a secondary-source approach was adopted instead.

5.0 Analysis and Discussion

This section explains how shadow variations can be linked to Earth’s axial tilt, solar declination and the Sun’s changing position in the sky.

5.1 Scientific Reasoning: Linking Observations to Astronomical Concepts

Our findings suggested that shadow cast by a fixed vertical object (gnomon) reaches its shortest length at solar noon, and that this shortest length varies from day to day and across seasons. These observations can be directly explained by the Sun’s apparent motion and Earth–Sun geometry.

At solar noon, the Sun crosses the local meridian and attains its maximum altitude angle above the horizon for that day. Because shadow length is inversely related to solar altitude, the Sun’s highest position naturally produces the shortest shadow. This confirms that solar noon is defined by the Sun’s position in the sky, not by clock time, which may differ due to time zones and the equation of time

However, the Sun’s altitude at solar noon is not the same throughout the year. This is due to the tilt of Earth’s rotational axis, which is approximately 23.5 degrees relative to its orbital plane. As Earth revolves around the Sun, this axial tilt causes the Sun’s apparent position, known as solar declination, to change gradually over the year. When the Sun’s declination is higher, the Sun appears higher in the sky at noon, producing a shorter shadow. When the declination is lower, the Sun appears lower, resulting in a longer noon shadow.

$$h = 90^\circ - |\phi - \delta|$$

Therefore, the variation in the shortest shadow length is a direct result of Earth's axial tilt and orbital motion, not changes in the stick or observation method. The shadow effectively acts as a natural indicator of the Sun's changing position in the sky.

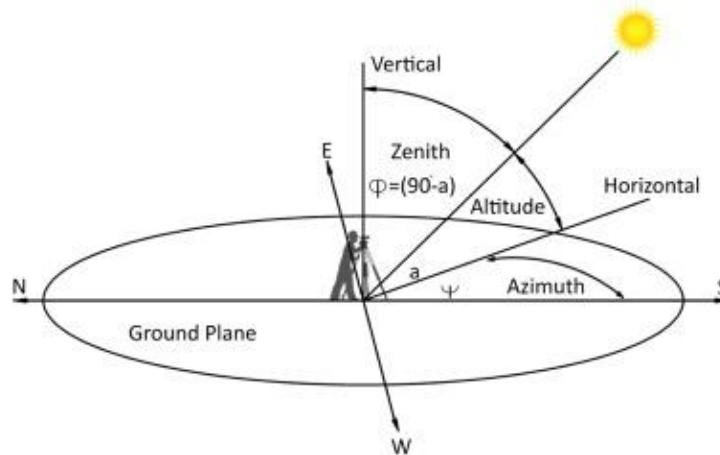


Figure 5.1: Solar altitude and shadow length at solar noon.

5.2 Addressing Common Misconceptions

One common misconception is that the Sun is always directly overhead at noon. In reality, the Sun is only directly overhead at specific locations between the Tropic of Cancer and the Tropic of Capricorn, and only on certain days of the year. For most locations on Earth, including Malaysia, the Sun is overhead only on limited dates and is usually slightly tilted even at solar noon.

Another widespread misunderstanding is that changes in shadow length are caused by variations in the distance between the Earth and the Sun. While Earth's orbit is slightly elliptical, the change in distance is relatively small and does not significantly affect shadow length. Instead, the dominant factor is the angle at which sunlight strikes Earth's surface, which is controlled by

Earth's axial tilt. This misconception is similar to the incorrect belief that seasons are caused by Earth being closer to or farther from the Sun.

By correcting these misconceptions, shadow observation can help students develop a more accurate understanding of how Earth's motion influences daily and seasonal changes.

5.3 Societal and Practical Implications of Shadow Patterns

Shadow patterns have played an important role in human society since ancient times. Early civilizations used shadows to design sundials, which relied on predictable shadow movement to measure time. These devices were carefully designed based on location and seasonal changes in the Sun's path.

In architecture, understanding shadow behaviour influenced the orientation of buildings, courtyards, and openings to control sunlight exposure and thermal comfort. Even today, modern building design uses solar and shadow analysis to improve energy efficiency and indoor comfort.

In contemporary applications, shadow patterns are especially important for solar energy systems. Solar panels are positioned based on expected solar angles throughout the year to maximise energy collection. Seasonal changes in the Sun's height must be considered to ensure consistent and efficient power generation.

These examples show that a simple phenomenon such as a changing shadow has meaningful connections to technology, sustainability, and daily life.

5.4 Overall Insight

The daily and seasonal variation of shadow length is a clear and observable consequence of Earth's motion in space. By linking simple field observations to Earth's axial tilt, solar declination, and apparent solar motion, this case study demonstrates how fundamental astronomical principles can be understood through everyday experiences. This highlights the value of field astronomy in connecting scientific theory with real-world observation.

6.0 Conclusion and Recommendations

This section presents the conclusion and recommendations based on the investigation of shadow length variation.

6.1 Conclusion

This investigation confirms that the daily and seasonal variation in shadow length is a direct consequence of Earth's axial tilt and its orbital motion around the Sun. The shadow produced by a fixed vertical object (gnomon) changes length because the Sun's altitude angle at a given clock time varies throughout the year. At solar noon, the Sun reaches its highest point in the sky for that day, producing the shortest shadow. However, the solar altitude at noon changes over the year due to the variation in solar declination caused by Earth's tilt of approximately 23.5° relative to its orbital plane. Therefore, even when the gnomon, location, and clock time remain constant, the shadow length differs on different dates. This demonstrates that shadows act as natural indicators of Earth–Sun geometry and provide observable evidence of the reasons behind seasonal changes.

6.2 Recommendation

To strengthen public astronomy education and promote deeper understanding of Earth–Sun relationships, schools and educational institutions should incorporate **more hands-on learning activities into their science curriculum**. One effective approach is to **install interactive sundial exhibits or gnomon stations** within school grounds. These installations would allow students to directly observe shadow movement and record measurements over time, helping them connect theoretical concepts such as solar declination and axial tilt with real-world observations. By engaging with a physical sundial, students can experience the changing shadow patterns throughout the day and across different seasons, making abstract astronomical ideas more tangible and memorable.

In addition to static exhibits, schools should **organize regular field astronomy sessions where students can conduct simple experiments outdoors**. These activities can include measuring shadow lengths at different times of the day, identifying solar noon, and comparing results across

different dates. Such experiences reinforce the scientific method by encouraging students to collect data, analyze trends, and draw conclusions based on evidence. Regular practice will also improve students' observational skills and promote curiosity about the natural world.

Furthermore, the integration of digital learning tools can enhance comprehension and accessibility. **Interactive apps or web-based simulations** that model shadow changes across various latitudes and seasons would allow students to visualize the effects of Earth's tilt and orbit. These tools can complement outdoor activities by providing immediate feedback and enabling exploration of scenarios that may be difficult to observe directly, such as shadow behavior in extreme latitudes or during solstices and equinoxes.

Finally, **community outreach programs and public demonstrations** should be promoted to raise awareness beyond the classroom. Science clubs, community centres, and museums can host **workshops on sundials, celestial navigation and seasonal shadow patterns**. Such initiatives can help the general public appreciate the relevance of basic astronomy in daily life and foster a culture of scientific literacy. Through a combination of interactive exhibits, hands-on activities, digital tools, and community engagement, astronomy education can become more engaging and impactful for students and the wider community.

7.0 Team Work

Group Member	Assigned Sections
Mohidul Hossain	1.0 Introduction 3.0 Relevant Astronomy Concepts
Tan Zhe Lam	2.0 Description of Issue 4.0 Investigation Method
Ahmed Fathi Salem	5.0 Analysis and Discussion
Tan Yi Ya	6.0 Conclusion & Recommendations Final Formatting

8.0 References

- Astronomical Almanac. (2024). *The astronomical almanac for the year 2024*. U.S. Government Publishing Office.
- Aveni, A. F. (2008). *People and the sky: Our ancestors and the cosmos*. Thames & Hudson.
- Berger, A. L. (1976). Obliquity and precession for the last 5,000,000 years. *Astronomy and Astrophysics*, 51(1), 127–135.
- Cutler, T. J. (2003). *Dutton's nautical navigation* (15th ed.). Naval Institute Press.
- Duffett-Smith, P., & Zwart, J. (2011). *Practical astronomy with your calculator or spreadsheet* (4th ed.). Cambridge University Press.
- Esenak, F. (2007). *Thousand year canon of solar eclipses: 1501 to 2500*. Astropixels Publishing.
- Green, R. M. (1985). *Spherical astronomy*. Cambridge University Press.
- Greenwood, M. (2022). Geometry and the Cause of the Seasons: The Changing Hours of Daylight and Elevation Angle of the Sun at Noon. *The Physics Teacher*. AIP Publishing. 10.1119/5.0038965
- Karttunen, H., Kröger, P., Oja, H., Poutanen, M., & Donner, K. J. (Eds.). (2017). *Fundamental astronomy* (6th ed.). Springer.
- Krupp, E. C. (2003). *Echoes of the ancient skies: The astronomy of lost civilizations*. Dover Publications.
- Lee, W. K., Nguyen, H., Kwak, Y.-S., & Kil, H. (2025). Seasonal Variation of Daytime Electron Density Irregularity Distributions in the Bottom and Top Sides of the Low Latitude F Region. *JGR Space Physics*. AGU. 10.1029/2025JA033902
- Mayall, R. N., & Mayall, M. W. (1994). *Sundials: Their construction and use* (3rd ed.). Sky Publishing Corporation.

- McCarthy, D. D., & Seidelmann, P. K. (2018). *Time: From Earth rotation to atomic physics* (2nd ed.). Cambridge University Press.
- Meeus, J. (1998). *Astronomical algorithms* (2nd ed.). Willmann-Bell.
- Milham, W. I. (1945). *Time and timekeepers*. Macmillan.
- Rohr, R. R. J. (1996). *Sundials: History, theory, and practice*. Dover Publications.
- Roy, A. E., & Clarke, D. (2003). *Astronomy: Principles and practice* (4th ed.). CRC Press.
- Smart, W. M. (1977). *Textbook on spherical astronomy* (6th ed.). Cambridge University Press.
- Waugh, A. (1973). *Sundials: Their theory and construction*. Dover Publications.
- Wang, Y.-K. (2017). Mathematical model for solving the relationship of shadow length changing over time through solar elevation. *Springer Nature*. Atlantis Press. 10.2991/amcce-17.2017.38
- Wolfson, R. (2020). *Essential university physics: Volume 1* (4th ed.). Pearson.